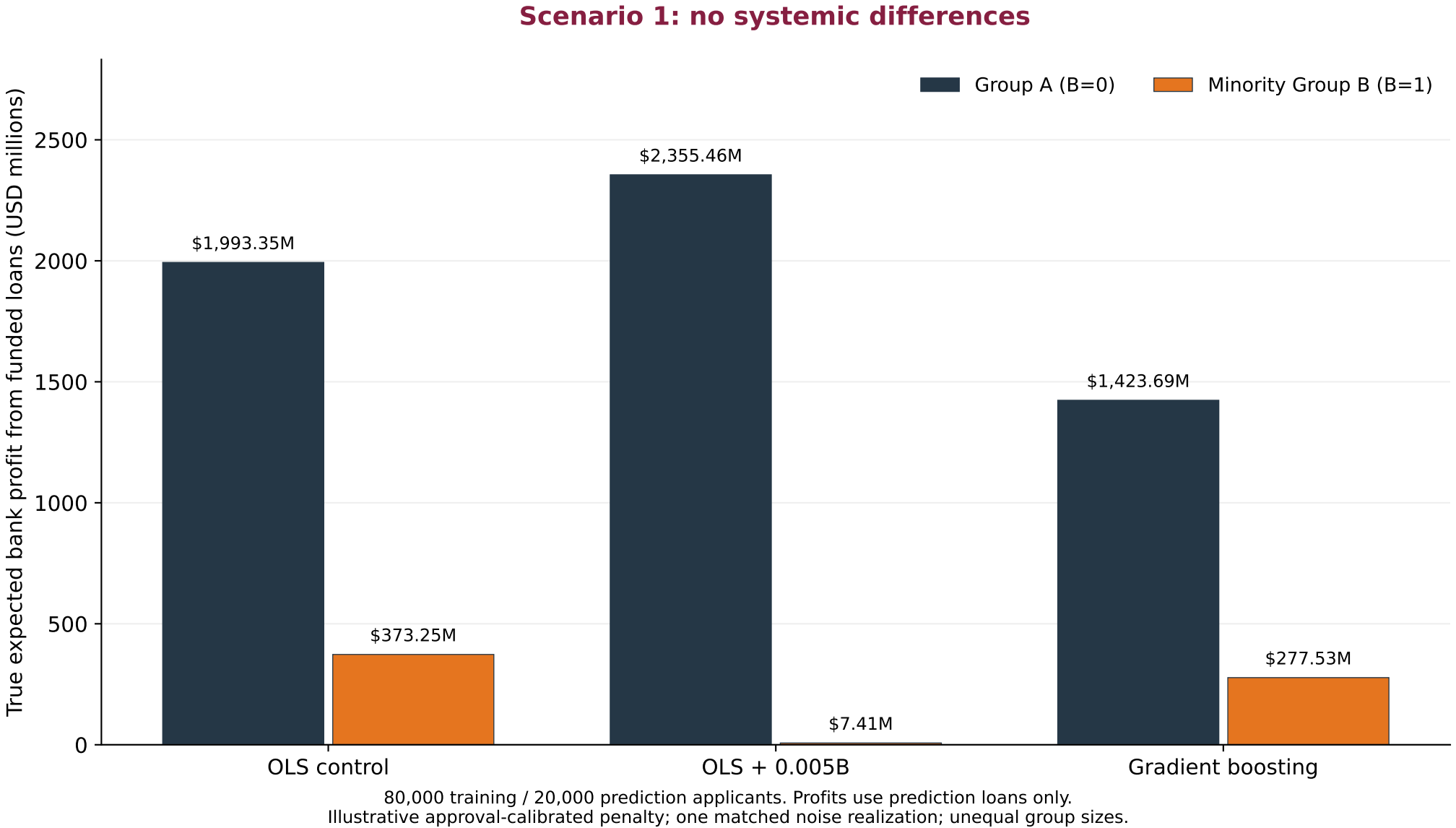
Scenario 1: no systemic differences
Group A (B=0)
Minority Group B (B=1)
True expected bank profit from funded loans (USD millions)
2500
2000
1500
1000
500
0
$1,993.35M
$373.25M
$2,355.46M
$7.41M
$1,423.69M
$277.53M
OLS control
OLS + 0.005B
Gradient boosting
80,000 training / 20,000 prediction applicants. Profits use prediction loans only.
Illustrative approval-calibrated penalty; one matched noise realization; unequal group sizes.